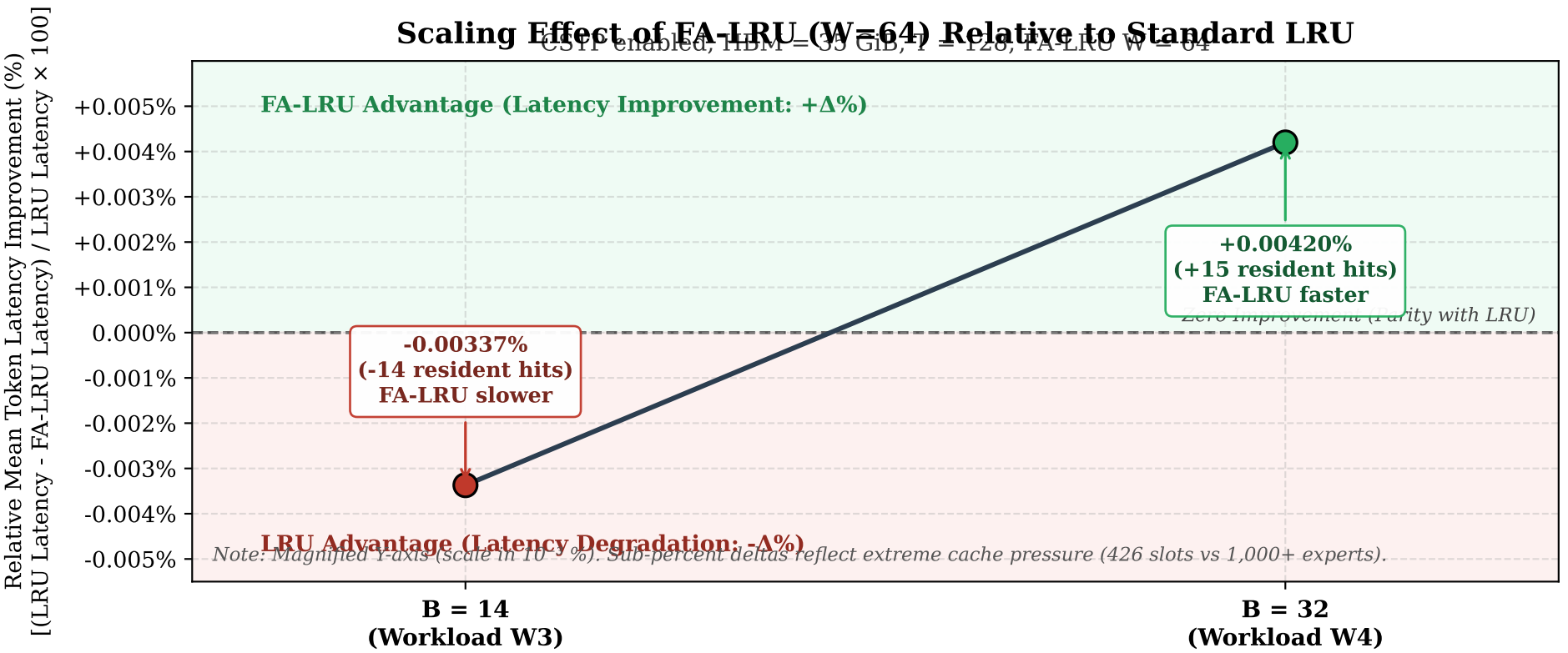


Supporting Table: Absolute Metrics and Deltas (CSTP Enabled, Repetition 1 & 2 Identical)

Metric	W3: LRU (B=14)	W3: FA-LRU (B=14)	W3 Delta (FA-LRU vs LRU)	W4: LRU (B=32)	W4: FA-LRU (B=32)	W4 Delta (FA-LRU vs LRU)
Coalesced Resid	1,657	1,643	**-14 (-0.84%)**	1,417	1,432	**+15 (+1.06%)**
Expert Reloads f	316,227	316,241	**+14 (+0.004%)**	536,034	536,019	**-15 (-0.003%)**
Total DDR5 Traff	29,217.32	29,218.20	**+0.88 (+0.003%)**	48,738.84	48,736.81	**-2.03 (-0.004%)**
Exposed Memory	447,698.51	447,713.80	**+15.30 (+0.003%)**	747,863.57	747,831.75	**-31.82 (-0.004%)**
Mean Token Late	3,565.37	3,565.49	**+0.12 (-0.00337%)**	5,950.06	5,949.81	**-0.25 (+0.00420%)**

Interpretation of Sign Inversion Across Concurrency Scaling:
- At lower batch concurrency (B=14, Workload W3), cross-prompt expert sharing is sparse. The 40-layer sequential working set displaces experts before repeated access occurs. Frequency protection in FA-LRU retains older lines with low re-use probability, producing minor cache churn (-14 resident hits, +0.88 GB DDR5 traffic, -0.00337% relative latency degradation).
- At higher batch concurrency (B=32, Workload W4), inter-sequence overlap expands substantially. FA-LRU leverages its bottom W=64 window to selectively evict unconsumed speculative prefetches (f=1) while protecting frequently requested demand lines (f=2). This targeted retention avoids 15 DDR5 expert reloads, reduces DDR5 traffic by 2.03 GB, and yields a net latency gain (+0.00420%).
- Determinism note: The two repetitions produce bit-exact identical values; differences represent exact architectural behavior.